Academic case study | Sophia Learning — Introduction to Networking
GreenLeaf Switching, VLANs & Layer 2 Security

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1. Switching Architecture

1a. Switch Roles

At the Portland location, PDX-CORE-SW01 is the core switch and acts as the central distribution point for the network. PDX-ACC-SW01, PDX-ACC-SW02, and PDX-ACC-SW-03 are access switches that connect end devices in different areas of the building. At the Eugene location, EUG-CORE-SW01 is the core switch, while EUG-ACC-SW01 and EUG-ACC-SW02 provide connections for office and warehouse devices. Traffic from an end device first reaches an access switch and then travels over a trunk link to the core switch, where it can be forwarded to another access switch or toward the site router.

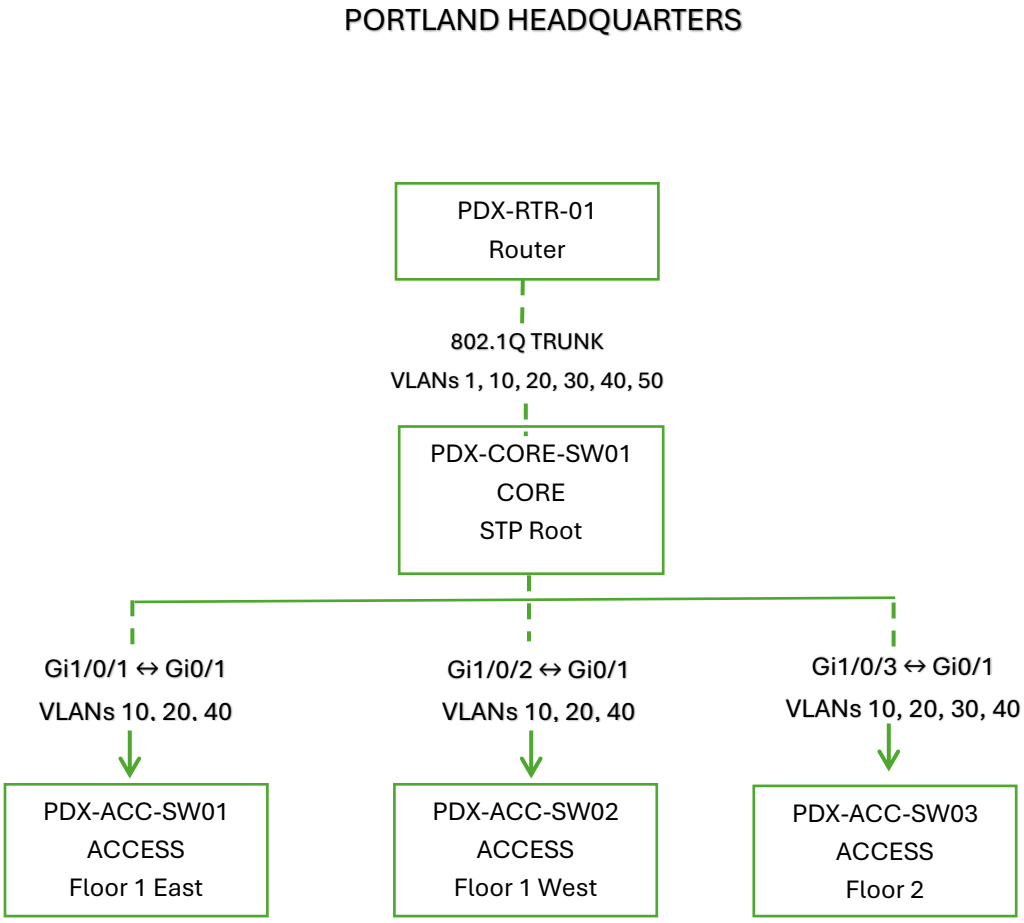


Figure 1. GreenLeaf Portland Layer 2 Switching and VLAN Topology. Adapted from the GreenLeaf Layer 2 Switch Topology Diagram.

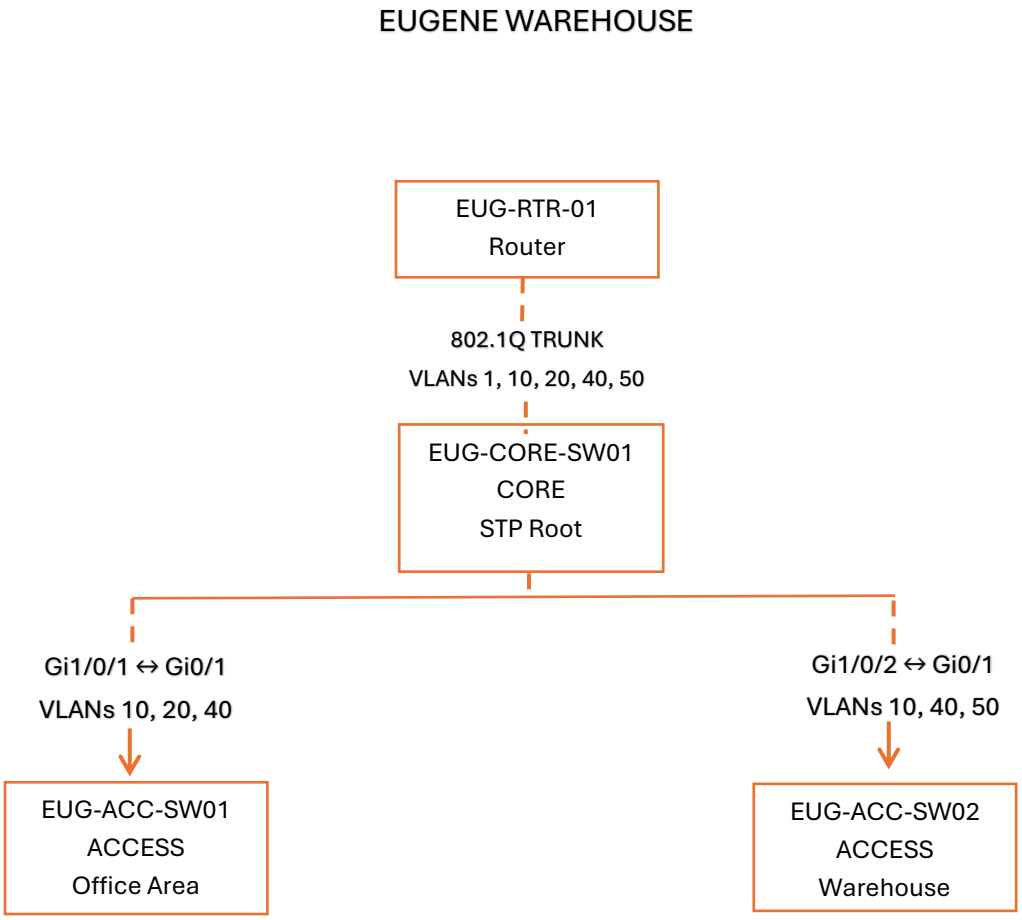


Figure 2. GreenLeaf Eugene Layer 2 Switching and VLAN Topology. Adapted from the GreenLeaf Layer 2 Switch Topology Diagram.

1b. Frame Forwarding and MAC Learning

A switch learns MAC addresses by looking at the source MAC address of each frame it receives and associating that address with the port where the frame entered. If the destination MAC address is already known, the switch forwards the frame only to the correct port. Broadcast frames, such as an ARP request, are forwarded to the other ports within the same VLAN, while an unknown-unicast frame is flooded within that VLAN until the destination is learned. For example, when a workstation on VLAN 10 sends traffic toward its gateway, the frame enters PDX-ACC-SW01 through an access port and is forwarded to PDX-CORE-SW01 over the Gi0/1 trunk. An 802.1Q VLAN 10 tag is added when the frame travels across the trunk so the switches know which VLAN it belongs to. STP keeps the Layer 2 path loop-free by selecting the core switch as the root bridge and determining which switch ports forward traffic.

2. VLAN Implementation

2a. VLAN Table

VLAN ID	VLAN Name	Purpose	Site(s)	IP Subnet
10	Users	Employee PCs	Both	10.x0.10.0/24–/25
1	Management	Device management	Both	10.x0.1.0/24
20	VoIP	IP Phones	Portland	10.10.20.0/24
30	Servers/Local	Servers/Services	Both	10.x0.30.0/25
40	Wireless	Corporate WiFi	Both	10.x0.40.0/24-/25
50	Guest/Warehouse	Guest/Inventory	Both	10.x0.50.0/24-/25
999	BLACKHOLE	Security	Both	None

2b. Encapsulation Summary

Interface	Mode (Access/Trunk)	VLAN(s)	Description
Gi0/1	**Trunk**	**1, 10, 20, 40**	**Trunk to PDX-CORE-SW01**
Fa0/1	**Access**	**10 data, 20 voice**	**User PC – Marketing Manager Office 101**
Fa0/21	**Access**	**10**	**PDX-PRINTER-01 – Marketing Printer**
Fa0/23	**Access**	**20**	**Conference Phone – Room 110**
Gi0/2	**Access**	**999**	**Unused – Reserved for Future Uplink**

2c. Inter-VLAN Routing Method

GreenLeaf uses router-on-a-stick to allow devices in different VLANs to communicate. Traffic is sent over an 802.1Q trunk to its site router, where separate router subinterfaces act as the default gateways and route traffic between the VLANs.

3. Switching Protocols and Security

3a. Portland Subnets

The STP root bridge at Portland is PDX-CORE-SW01, while EUG-CORE-SW01 is the root bridge at Eugene. On PDX-ACC-SW01, Gi0/1 is the root port that connects toward the Portland core switch, while Fa0/1 is an example of a designated forwarding port. GreenLeaf uses Rapid-PVST+, and the core switches have a lower priority of 4096 so they are elected as the root bridges at each site. The current hub-and-spoke design has no blocked ports because there are no redundant Layer 2 paths creating loops.

3b. Port Security and Protections

GreenLeaf uses port security on access ports to limit which devices can connect. On PDX-ACC-SW01, user ports such as Fa0/1 allow a maximum of three MAC addresses and use resrtict mode if a violation occurs, while ports such as Fa0/2 and Fa0/23 use sticky MAC learning, allow only one MAC address, and shut down the port if a violation occurs. PortFast and BPDU Guard are enabled on end-device ports so devices can begin forwarding quickly while still protecting the network from unexpected BPDUs. Trunk links are hardened by using VLAN 999 (BLACKHOLE) as the native VLAN and by allowing only the VLANs that are needed on each trunk. On the Portland core switch, root guard is also enabled on the downlinks to the access switches to help prevent an access switch from becoming the STP root.

4. Physical Layer Note

GreenLeaf uses Cat6 UTP for horizontal cabling at the Portland headquarters. It supports 1000BASE-T Gigabit Ethernet with a maximum distance of 100 meters (328 feet). The backbone connections between the Portland IDFs and MDF use Cat6A F/UTP cabling, which supports 10GBASE-T and has a maximum distance of 100 meters. Cat6A was selected so the backbone can support a future 10 Gigabit upgrade without replacing the cabling. At the Eugene warehouse, industrial-rated Cat6 UTP is used on the warehouse floor. It has an oil-resistant jacket, an extended temperature range, and is plenum-rated for open ceiling areas. The longest documented Portland horizontal run is 78 meters, the Portland backbone links are 25 and 30 meters, and the longest Eugene warehouse run in 92 meters, so all documented runs remain within the 100-meter limit.